

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF ENVIRONMENTAL AND GEOGRAPHICAL SCIENCE

EGS3012S: ATMOSPHERIC SCIENCE

EXAMINATION: 2022

COURSE CONVENER: Prof. BJ Abiodun

EXTERNAL MODERATOR: Dr CD Hansen

TIME: 3 HOURS

TOTAL MARKS: 180

INSTRUCTIONS: This paper consists of 3 sections over 16 pages.

Answer **one** question from **SECTION A**.

Answer **four** questions from **SECTION B**.

Answer **all** questions from **SECTION C**.

Questions do **not** carry equal marks.

Budget your time (~1 mark/minute)!

Use clear, properly labelled, sketches and diagrams where appropriate.

No books, notes or other aids may be used in this examination.

An equation sheet is provided; use as appropriate.

Write the question numbers of SECTIONS A & B on the front of your answer booklet.

SECTION A

Answer **ONE** of the following essay questions (1 x 60 = **60 marks**)

(Read and answer the questions, be selective of your theory, distill the essentials to answer the question, don't just brain-dump. Think: how would I explain this to the person next to me? Give short but convincing answers.)

1. (a) The climate system responds to complex interactions between terrestrial, marine, and atmospheric components. Identify the key components of the climate system and discuss how the global energy balance is modulated by these, with attention to the role of water (in its gaseous, liquid, and solid phases).

Or

(b) Discuss the roles of the atmosphere in making the earth habitable and in making some parts of the world more habitable than others. Use diagrams, explanations, and examples of relevant atmospheric processes to illustrate your answer. Consider questions such as what if the atmosphere were completely transparent to infrared radiation, and how would the surface net radiation budget be different?

Or

(c) Given that General Circulation Models (GCMs) are currently the most viable technique for assessing a global system response to increased levels of CO₂ in the atmosphere, discuss the principal shortcomings and advantages of GCMs. In your discussion, give due consideration to GCMs formulation, underlying assumptions, types, applications, and usefulness in regional weather and climate prediction.

Or

Discuss if global warming is real and if climate mitigation, carbon dioxide removal, and solar radiation management can stop global warming. In your discussion, mention relevant concepts (e.g. radiative energy climate forcing, climate sensitivity, climate feedback, climate variability, and climate change), and how they relate to the warming. Also, discuss the advantages and disadvantage of each complementary and alternative approach mentioned above, and which of them you would support and why?

SECTION B

Answer **any four** questions (4 x 15 = 60 marks)

Some useful equations and constants (in case you need them).

Note that all symbols have their usual meanings as indicated in the lecture notes

$$\begin{aligned} G &= \left| \frac{g}{f_c} \cdot \frac{\Delta z}{\Delta d} \right| & U_g &= -\frac{|g|}{f_c} \cdot \frac{\Delta z}{\Delta y} & V_g &= +\frac{|g|}{f_c} \cdot \frac{\Delta z}{\Delta x} \\ V_T &= V_{g(2)} - V_{g(1)} & &= \frac{g}{f_c} \cdot \frac{\Delta(z_2 - z_1)}{\Delta n} & &= \frac{R}{f_c} \cdot \frac{\Delta T}{\Delta n} \left(\ln \left(\frac{P_1}{P_2} \right) \right) \\ u_T &= u_{g(2)} - u_{g(1)} & &= -\frac{g}{f_c} \cdot \frac{\Delta(z_2 - z_1)}{\Delta y} & &= -\frac{R}{f_c} \cdot \frac{\Delta T}{\Delta y} \left(\ln \left(\frac{P_1}{P_2} \right) \right) \\ v_T &= v_{g(2)} - v_{g(1)} & &= \frac{g}{f_c} \cdot \frac{\Delta(z_2 - z_1)}{\Delta x} & &= \frac{R}{f_c} \cdot \frac{\Delta T}{\Delta x} \left(\ln \left(\frac{P_1}{P_2} \right) \right) \\ \zeta_r &= \frac{\Delta V}{\Delta x} - \frac{\Delta U}{\Delta y} & D &= \frac{\Delta U}{\Delta x} + \frac{\Delta V}{\Delta y} & f_c &= 2\Omega \sin(\phi) \end{aligned}$$

Constants:

$$R = 287 \text{ J kg}^{-1} \text{ K}^{-1} \quad |g| = 9.8 \text{ m s}^{-2} \quad \Omega = 0.7292116 \times 10^{-4} \text{ radians s}^{-1}$$

2. (a) In terms of the continuity equation, describe five (5) processes that give rise to convergence and divergence in the atmosphere. [10 marks]
- (b) The following wind data were received from 50 km to the east, north, west, and south of Johannesburg respectively: [10 m s⁻¹, 180°], [10 m s⁻¹, 90°], [10 m s⁻¹, 0°], [10 m s⁻¹, 270°]. Calculate the approximate horizontal divergence. [5 marks]
3. (a) What is thermal wind? [3 marks].
- (b) Use **Figure 1** to calculate the following winds over Cape Town.
- (i) the geostrophic wind at 850hPa [4 marks]
 - (ii) the thermal wind in the layer between 850hPa and 500hPa [4 marks]
 - (iii) geostrophic wind at 500hPa [4 marks]

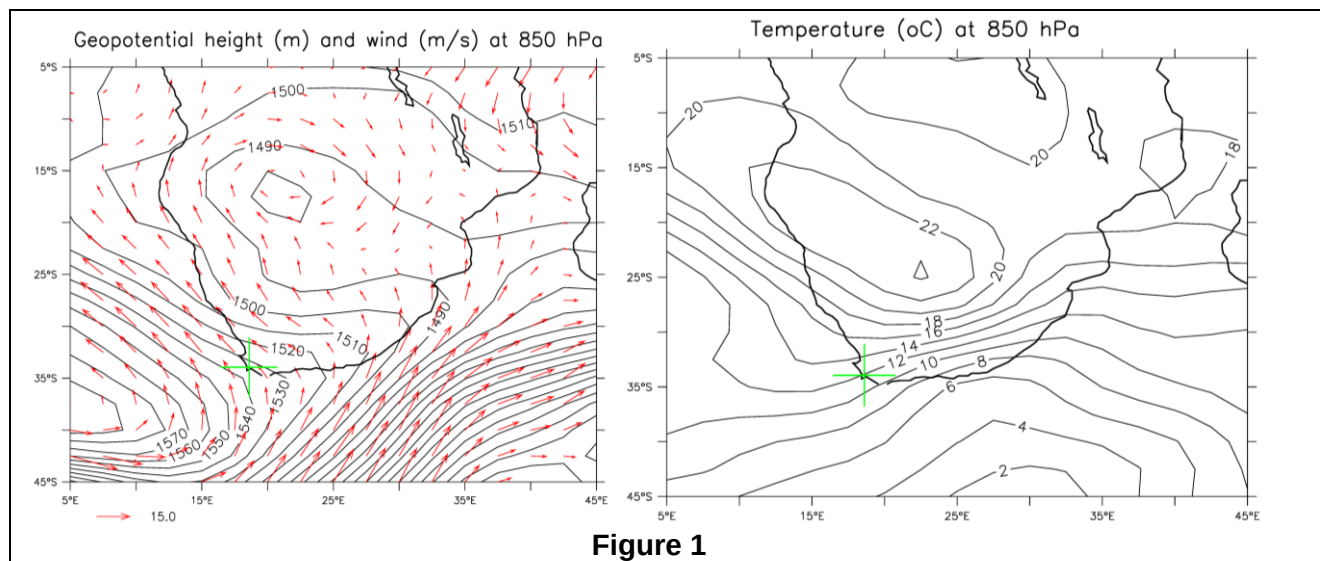


Figure 1

4. (a) What is the Bergeron-Findeison theory? Explain how it enables systems such as thunderstorms to form and precipitate in such a short period of time. **[10 marks]**

(b) Using the Skew-T plot rules, complete the table below: **[5 marks]**

P (hPa)	T (°C)	T _w (°C)	T _d (°C)	x (g/kg)	x _s (g/kg)	LCL (hPa)	θ (°C)
825	6.0	4.0					

5. With the aid of diagrams, outline the similarities and differences between the following atmospheric features.

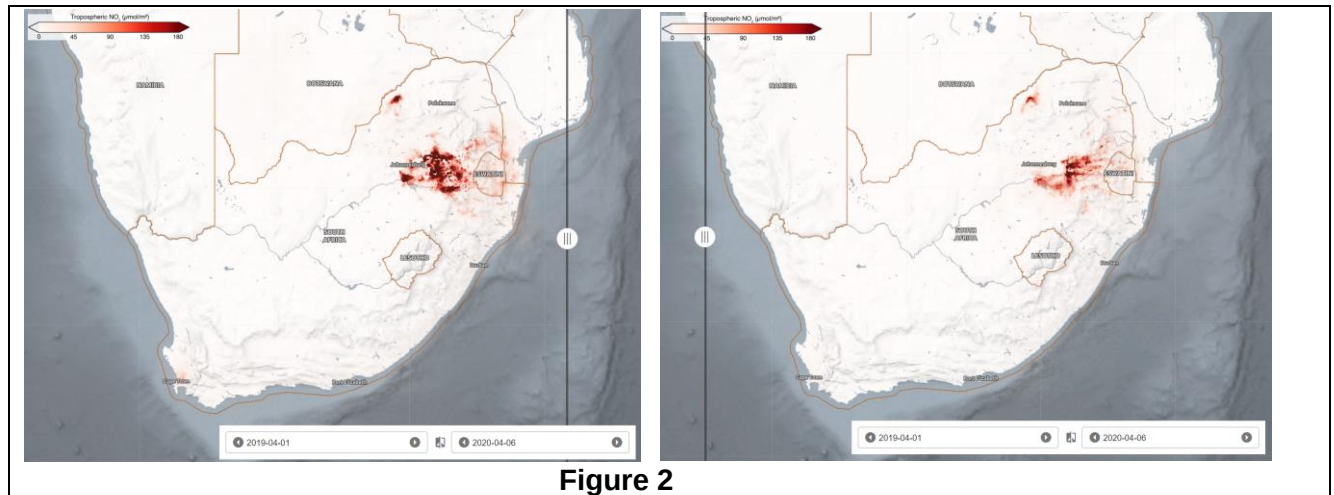
- Climate variability and climate change **(3 marks)**
- Sea-breeze and monsoon flow **(3 marks)**
- Tornado and cut-off low **(3 marks)**
- Tropical and Mid-latitude atmospheric dynamics **(3 marks)**
- Warm Front and Cold Front **(3 marks)**

6. With the aid of diagrams, describe the anatomy of a hurricane. What would the traces of temperature, pressure, and winds look like in an E-W cross section through the centre of the hurricane? What ingredients are necessary for hurricane formation? What factors inhibit hurricane formation? **[15 marks]**

7. (a) Use **Figure 2** to answer this question. COVID-19 provided a natural experiment on the impacts of certain human activities on air pollution. The left panel of Figure 2 is a satellite image of NO₂ concentrations on 1 April 2019, and the right panel is 1 April 2020.

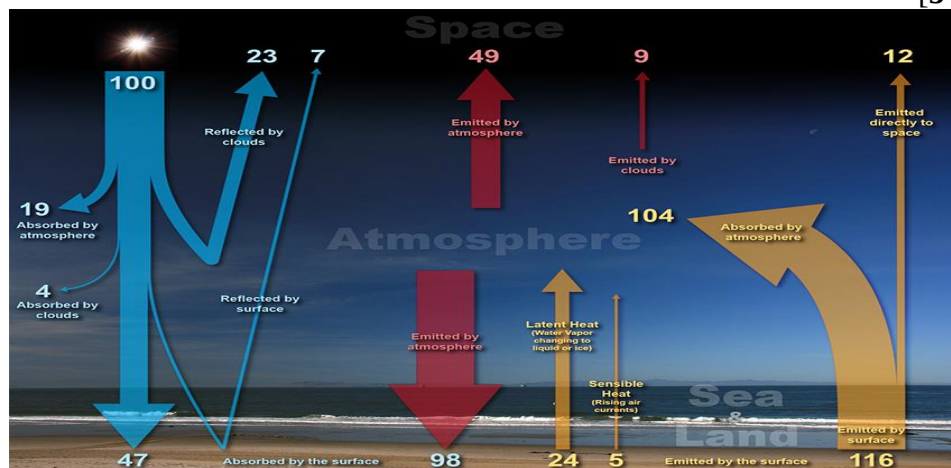
- What spatial differences do you note in the NO₂ concentrations around South Africa?
- Why are there changes in NO₂ concentrations during lockdown in some areas of the country but not others?
- What does that tell you about the drivers of NO₂ pollution in those areas?

[10 marks]



- (b) Use the information in **Figure 3** to generate a table for radiation and heat budget
- at earth's surface
 - in the atmosphere and
 - at the top of the atmosphere.

[5 marks]



8. (a) **Figure 4** is a synoptic chart for the 22nd of a particular month in the year 2011 at time 00:00Z.....this day is Day 0. Detach the last page of this paper and submit this together with your answer booklet.

The column Day +1 is 24 hours later, (i.e. the 23rd) at 00Z. Using your knowledge of South African weather systems draw in isobars of the **500hPa height** in the upper row for Day 0 and Day +1 and label regions of high and low pressure. In the bottom row draw in **sea level pressure** isobars for Day +1 and label regions of high and low pressure. On the **day 0 slp map** label the synoptic systems present. [10 marks]

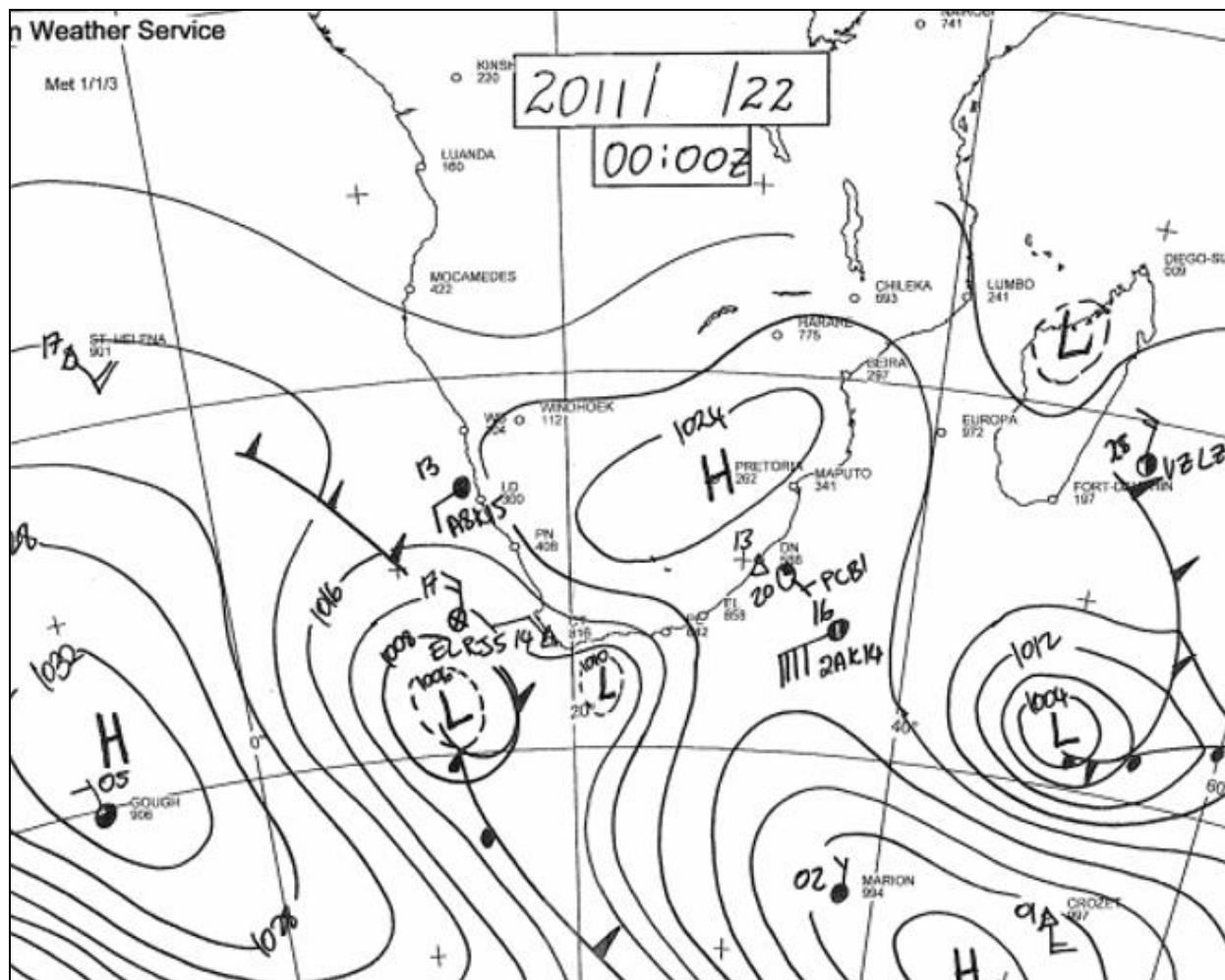


Figure 4

	Day 0		Day +1	
	Johannesburg	Cape Town	Johannesburg	Cape Town
Wind speed				
Wind direction				
Rain (Yes/No)				
If Yes, type of rain				
Synoptic system affecting the city				

(b) The table above lists a set of variables and two cities for Day 0 and Day +1. Fill in values for each variable that would likely be recorded at the airport of each city on each day. What is the dominant synoptic circulation type controlling the weather over each city? **[5 marks]**

SECTION C
Answer ALL questions (60 marks)

S/N	Question	Answer
1	A low pressure on a constant height chart corresponds to a _____ on a constant pressure chart (a) low pressure. (b) low height. (c) high pressure. (d) high height.	
2	A rising parcel of air that does not exchange heat with its surroundings is referred to as (a) isothermal ascent. (b) an adiabatic process. (c) forced lifting. (d) advection.	
3	A supercell storm consists of a single rotating updraft. True or False?	
4	A supercell storm is: (a) a thunderstorm that produces several tornadoes (b) a thunderstorm that produces a category F-5 tornado (c) a thunderstorm with an extremely large downburst (d) an enormous thunderstorm that lasts for several hours	
5	A temperature gradient results in (a) a thermal wind, which in a cold cored surface low causes decreasing wind velocity with height. (b) a thickness gradient and thermal wind, which always opposes geostrophic flow (c) a thickness gradient and Ekman spiral (d) b + c (e) none of the above	
6	A typical sequence associated with the passage of a cold front over Cape Town is: (a) Cut-off low, cold front, southerly meridional flow, ridging south Atlantic high (b) Coastal low, cold front, southerly meridional flow, ridging south Atlantic high (c) Coastal low, tropical cyclone, southerly meridional flow, ridging south Atlantic high (d) Tropical low, cold front, tropical temperate trough, ridging south Atlantic high (e) Low pressure ridge, coastally trapped wave, baroclinic instability, elevated inversion (f) a+e	
7	Air pollutant emissions can be from which of the following sources, except : (a) Natural	

	(b) Anthropogenic (c) Gas (d) Stationary (e) None of these	
8	Anti-cyclonic gradient winds are super-geostrophic due to the centrifugal forces being additive to the pressure gradient force. True or False?	
9	As the difference between the dry-bulb temperature and wet-bulb temperature increases, the relative humidity: (a) increases (b) remains constant at a value less than 100% (c) decreases (d) remains constant and equal to 100%	
10	Assume you are visiting a friend in Darwin, on the north shore of Australia, and a hurricane approaches from the north. The highest storm surge should be from the (a) east (b) west (c) north (d) south	
11	At 25°C, the atmosphere is saturated with water vapour. If the temperature is increased to 30°C, with no addition of water vapour, one may conclude that the dew point temperature is: (a) 15°C (b) 20 °C (c) 30°C (d) 25°C (e) 50°C	
12	Atmospheric longwaves (i.e., Rossby waves) usually move from _____ at a speed that is _____ than that of the shortwaves. (a) east to west, faster (b) west to east, faster (c) east to west, slower (d) west to east, slower	
13	The Bergeron-Findeisen process will operate in a cloud with the ambient temperature of 3°C? True or False?	
14	Choose the most correct statement: (a) The wind shifts clockwise with the passage of a mid-latitude cyclone in the southern hemisphere. (b) Southerly meridional flow is associated with sharp drops in temperature over most of South Africa, especially the southern regions. (c) The summer thermal trough over the interior of South Africa is a baroclinic feature. (d) Easterly waves are baroclinic disturbances. (e) Cut-off lows are caused by a deepening of the easterly wave trough.	
15	Climate change is just one aspect of global warming.	

	True or False?	
16	Coastal lows are a cold cored cyclone. True or False?	
17	Daily variations in the weather of tropical regions are caused by the influence of different air masses. True or False?	
18	Deep Convection: (a) typically transfers heat downward during the day. (b) plays no significant role at night. (c) helps transfer heat throughout the atmosphere.	
19	Emission inventories are useful for which of the following, except (a) Establishing regulations (b) Health risk assessments (c) Establishing monitoring strategies (d) Air quality modeling (e) None of the above	
20	Frost forms when (a) the air temperature reaches the dew point. (b) the dew point is 0°C or below. (c) Water vapour changes into ice. (d) all of the above (e) only (b) and (c).	
20	If an air parcel is given a small push upward and it falls back to its original position, the atmosphere is said to be (a) stable. (b) unstable. (c) isothermal. (d) neutral. (e) adiabatic.	
20	If I run a Global Circulation Model (GCM) at horizontal resolution of 50 km resolution and a Regional Climate Model (RCM) at horizontal resolution of 100 km, which of the models will be appropriate for simulating a 300 km wide hurricane? (a) GCM (b) RCM (c) both models (d) none of the models	
21	In a multi-cell storm, each of the cells is always in the same stage of development. True or False?	
22	Hurricane winds rotate in an anticlockwise direction (a) in the Northern Hemisphere only. (b) in the Southern Hemisphere only. (c) in both the Northern and Southern Hemispheres. (d) in neither hemisphere.	

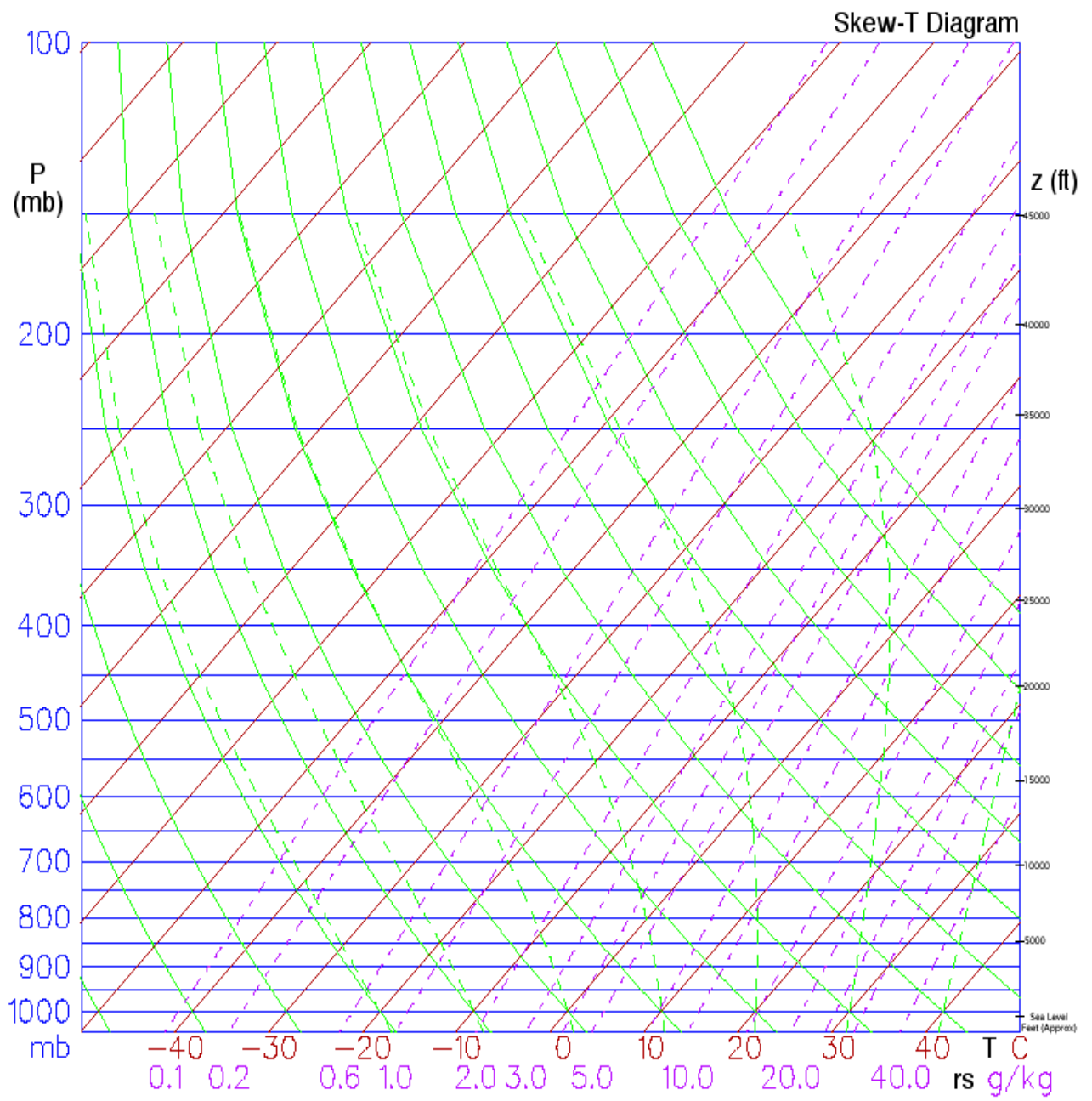
22	Hurricanes decrease in intensity over land because: (a) of friction (b) of lack of sufficient moisture (c) the eye collapses (d) a + b (e) b + c (f) a + c	
23	Global Climate Models... (a) can accurately simulate the climate 5 years into the future, but not 30 years into the future. (b) include direct simulation of cloud microphysics and the interaction of radiation with cloud individual particles. (c) are based on coupled sub-models of the components of the climate system, and realistically simulate the 20th century climate. (d) can realistically simulate nearly all the regional climate changes of the globe but fail to capture the projected polar warming.	
25	Hurricanes gain most of their energy from (a) Rossby waves. (b) insolation on their extensive, high-level clouds covers. (c) the strong winds that develop within them. (d) cold ocean water and latent heat release through evaporation. (e) none of the above.	
26	Ice-albedo feedback is a good example of negative feedback in climate system. True or False?	
28	Hurricanes can't form along the equator because (a) there isn't much water along the equator, it's mostly land. (b) the Coriolis force is too small along the equator. (c) there are no feedback mechanisms along the equator. (d) it's too humid for the ocean water to evaporate.	
29	If in the Southern Hemisphere at your home the surface wind is blowing from northeast, then the region of lowest pressure will be to the _____ of your home. (a) North (b) South (c) East (d) West	
30	If the earth rotates in the opposite direction with a higher angular speed, but the pressure gradient force remains the same, the speed of geostrophic wind would (a) show no change. (b) increase and blow in the opposite direction. (c) decrease and blow in the opposite direction. (d) increase and blow in the same direction. (e) decrease and blow in the same direction.	
31	In the cyclogenesis of a tropical cyclone, a tropical depression stage occurs between tropical storm and Hurricane Category 1. True or False?	

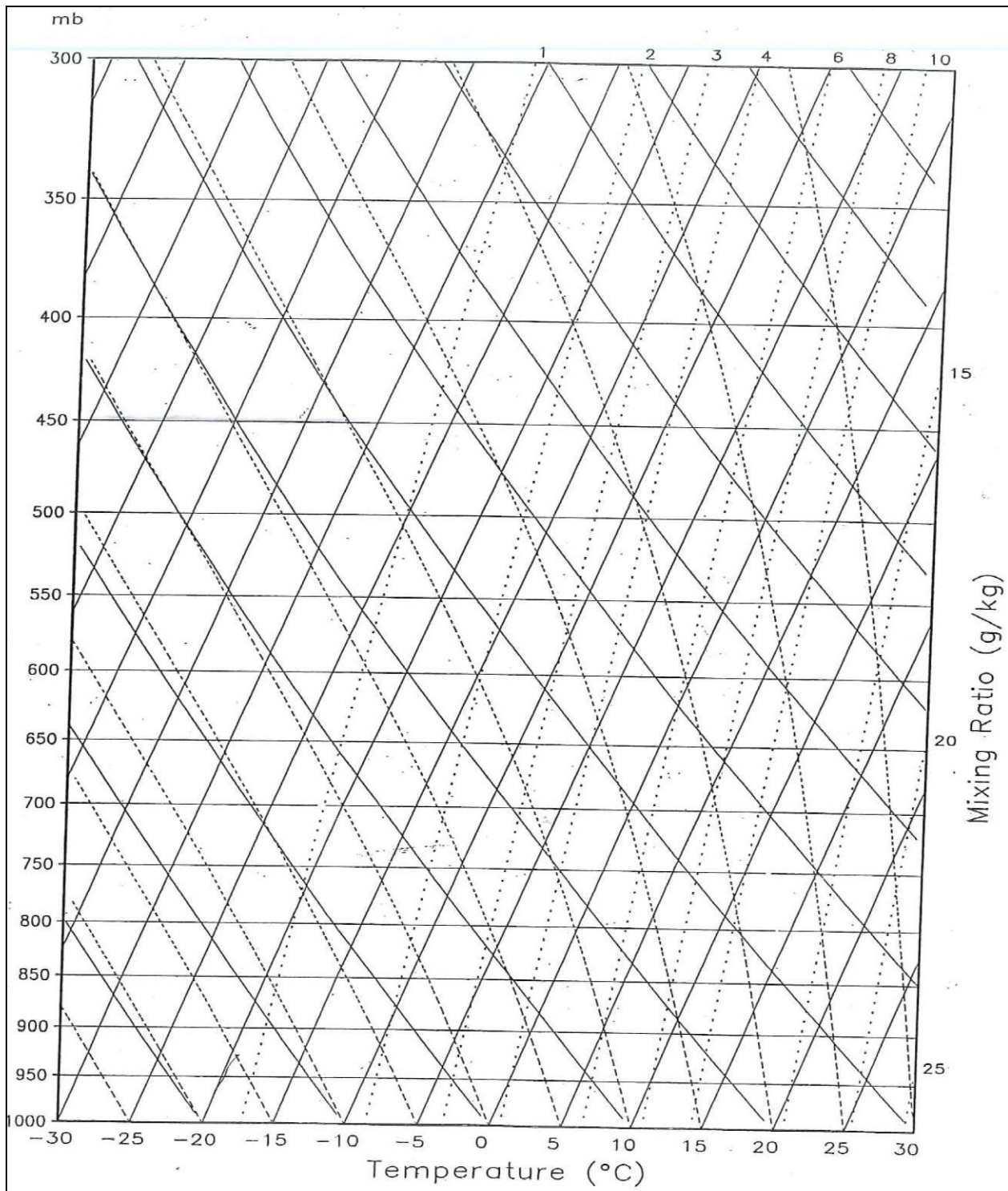
32	Both Northern and Southern Hemisphere cyclonic vorticity is associated with convergence and increase of relative vorticity with time. True or False?	
34	In the Southern Hemisphere, hurricanes and middle latitude cyclones are similar in that both (a) have surface weather fronts. (b) have wind intensifying with height above the ground. (c) will generally move from east to west. (d) all of the above. (e) none of the above.	
35	It is now established that global warming is due to the superimposition of several natural oscillations in the earth-atmosphere-ocean system at various time scales. True or False?	
36	Most hurricanes are associated with warm fronts. True or False?	
37	Most of the severe convection and earth's weather activities occur in the tropopause. True or False?	
38	On an upper-level chart the wind tends to blow (a) at right angles to the contours or isobars. (b) at an angle of 45° to the contours or isobars. (c) at an angle of 30° to the contours or isobars. (d) at an angle between 15° and 25° to the contours or isobars. (e) parallel to the contours or isobars.	
39	On an upper-level chart where the isotherms cross the isobars (or contours), the atmosphere is called (a) barotropic (b) geostrophic (c) hydrostatic (d) baroclinic	
40	On the Sariff-Simpson scale (a) a Category 5 hurricane has relatively moderate wind. (b) there is no attempt to estimate the storm surge height. (c) the surface pressure of the hurricane decreases as the category number increases. (d) there are seven different hurricane categories. (e) all of the above.	
41	The coriolis parameter is an important component of the cyclostrophic wind. True or False?	
42	The geostrophic wind is dependent only on the slope (gradient) of isobars. True or False?	
43	The gradient wind is an imbalance between pressure gradient force, coriolis force and centrifugal force. True or False?	

43	This prevents wind from following the direction of the horizontal pressure gradient force: (a) Friction with the ground. (b) The Earth's magnetic field. (c) Interaction with the solar wind. (d) The Coriolis effect. (e) All of the above.	
45	The pressure gradient force (a) can be measures by knowing the sea level pressure of just one point. (b) creates wind if it is horizontal. (c) will not necessarily results in movement of air. (d) is unrelated to the speed of air movement.	
46	The pressure gradient force is directed from lower pressure to higher pressure (a) at all places on the earth except at the equator. (b) only at the Northern Hemisphere. (c) only at the Southern Hemisphere. (d) at all places on the earth. (e) Never	
47	The vertical structure of a hurricane shows a _____ at upper-level and a _____ at the surface. (a) high pressure, high pressure (b) high pressure, low pressure (c) low pressure, high pressure (d) low pressure, low pressure	
48	The Jet stream (a) is an upper-level feature. (b) has a speed of about 50 km/hour. (c) occurs in regions of strong thermal wind component. (d) all of the above.	
49	Vehicle emissions are major primary sources of which of the following pollutants? (a) O₃ (b) PM₁₀ (c) CO (d) NO₂ (e) c+d (f) a+c (g) b+d (h) None of the above	
50	When upper-level divergence of air above a surface low pressure area is stronger than the convergence of surface air, the surface pressure will _____ and the storm itself will _____. (a) increase, intensify (b) increase, dissipate (c) decrease, intensify (d) decrease, dissipate	
51	Which of the following does not describe how water droplets grow to raindrops?	

	(a) Condensation (b) Aggregation (c) Collision (d) Coalescence	
52	Which of the following is a good example of Geoengineering Carbon Dioxide Removal strategy. (a) Stratospheric injection of aerosol (b) Biochar and afforestation (c) Space reflector (d) Desert albedo modification (e) All of the above	
53	Which of these is necessary for precipitation to occur over any city? (a) Passage of a cold front over the city (b) Vertical lifting of low-level moist air over the city (c) Landfall of tropical cyclone over the city (d) Orographic lifting of moist air over the city (e) All of the above	
54	Which of the following is not correct about thermal wind? (a) Low temperature occurs to the right of thermal wind in the Southern Hemisphere, and to the left of thermal wind in the Northern Hemisphere (b) Westerlies increase with height as thermal wind is additive to the geostrophic wind. (c) Easterly winds are weaker with height as they are opposed by thermal wind. (d) Cold cored cyclones intensify with height and are deep. (e) All of the above (f) None of the above	
55	The following pollutants impact negatively on both human health and ecosystems, except (a) CO (b) O₃ (c) SO₂ (d) NO₂ (e) None of the above	
56	The following species are anthropogenic gas-phase pollutant species, except (a) SO₂ (b) NaCl (c) NH₃ (d) CO (e) None of the above	
57	Which of the following is not an example of Geoengineering Solar Radiation Management strategy? (a) Stratospheric injection of aerosol (b) Biochar and afforestation (c) Space reflector (d) Desert albedo modification	

	(e) All of the above	
58	Which of these is not a mid-latitude weather system? (a) Thunderstorm (b) Squall line (c) Mesoscale convective system (d) Cutoff-low (e) Typhoon	
59	Which of these would you always expect during the passage of a gust front? (a) Gusty wind (b) Rising in temperature (c) Rotating wind (d) Drop in surface pressure	
60	Which would you expect to observe as the eye of a hurricane passes directly over your area? (a) An increase in surface temperature. (b) A very low surface pressure reading. (c) Very weak winds. (d) Little or no precipitation. (e) All of the above. (f) None of the above.	





Answer Question 8 on this page. Detach this page and submit it together with your answer booklet.

